MIS 505: Data Visualization
Final Project (Module 7)
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# Report on Global Greenhouse Gas Emissions (1950 to 2024)

## Overview

This summary report demonstrates how global level and country level greenhouse gas (Ghg) emission patterns changed from 1950 to 2024 using four dashboards that include ten charts and two tables. These dashboards and their accompanying charts and tables cover 74 years of emissions data across more than 190 countries. Together, they provide observations of long-term global trends, geographic distributions, differences among countries, and the largest increases and decreases in emissions over the time span.

One chart was created, and companion tables were considered, but not created or used, in the preparation of the dashboards and this report. This unused data visualization chart will be addressed in the section for Dashboard 4. Additionally, where countries returned null values for emissions values in the preparation of charts and tables, such as the top and bottom 20 countries by emissions, those countries were not considered in the analysis.

## Data Source

For this project, the dataset was obtained from Kaggle.com via the link below:

https://www.kaggle.com/datasets/lucalullo/global-ghg-gas-emissions-by-country-1950-2024?resource=download

The dataset is published by Our World in Data, which integrates several sources of information widely used in climate research. The dataset contained both total greenhouse gas emissions and greenhouse gas emissions per capita for more than 190 countries over the 74-year time span from 1950 to 2024.

While searching for a dataset to utilize for this project, I downloaded two others before setting on this one. The rejected datasets did not have time or dates as part of the data, and this one did, and I am more familiar with performing analyses with time series data. Additionally, I initially saved this downloaded dataset as a csv file, but Tableau did not recognize it until I converted the downloaded dataset to an Excel file format.

## Dashboard 1 — Diverging Storylines

### Purpose

To show the 74-year trajectory of global emissions and to distinguish between total emissions and per-capita emissions patterns.

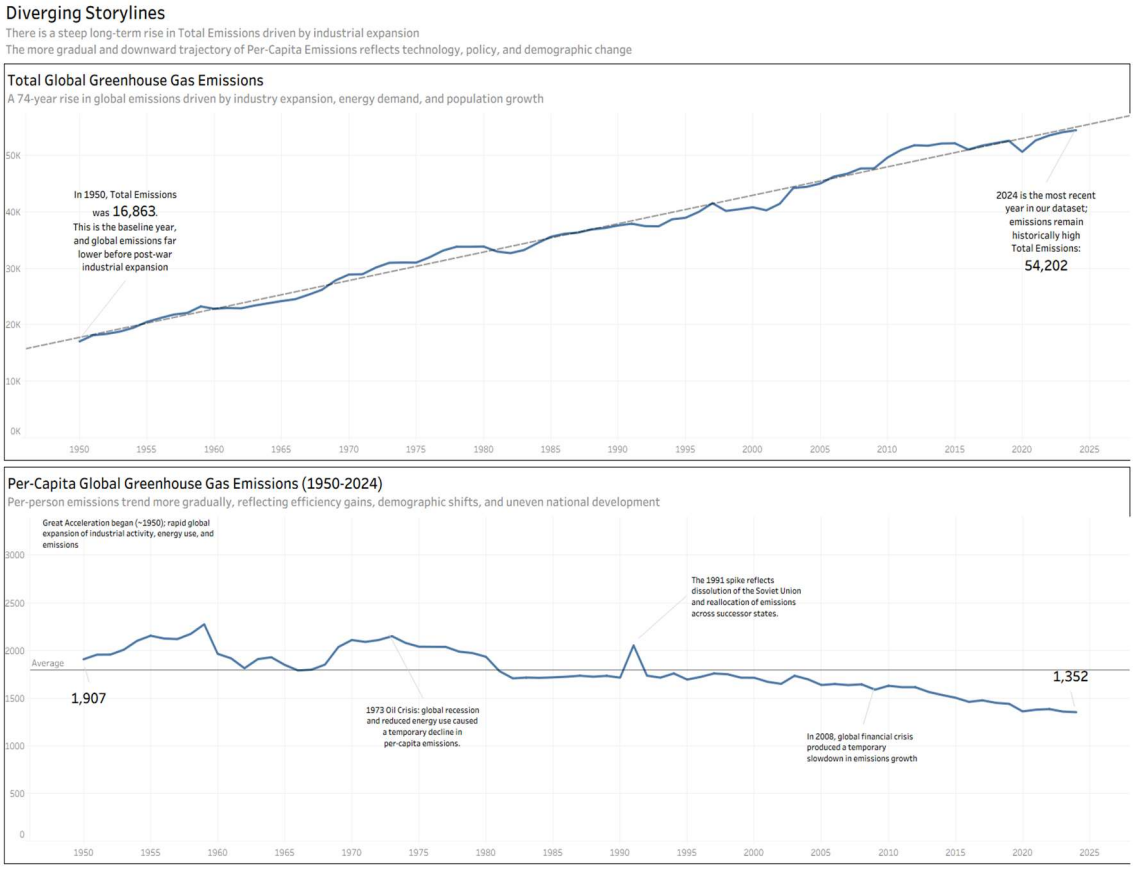
### Charts

1. **Total global greenhouse gas emissions rose over the 74-year period from 1950 to 2024.** The trajectory of the global total Ghg trendline is consistently upward. This reflects population growth, industrialization, and their resulting expansion of energy demands.
2. **Per-capita emissions follow a different pattern.** The per-capita Ghg emissions line chart shows moderate decline overall.

### Interpretation

The dashboard shows the divergence between the global total emissions, which grew substantially, and the per-capita emissions, where we can see both individual level contributions and their impact. The per-capita line and data suggest that population growth is a major factor in total emissions, and that efficiency gains, technology shifts, and policy interventions partially offset per-person emissions in some regions.

**Figure 1.**




### Effectiveness of Visuals

The two (2) line charts that make up Dashboard 1 clearly separate total emissions from per-capita emissions. This helps show how population growth and individual country's energy use contribute differently to global trends in greenhouse gas emissions. The divergence is immediately visible. As the total emissions trendline uses larger data, it is easier to spot changes in the trend on the per-capita line, as the changes are smaller, so I included several annotations to call out several specific historical points during the time span on that line chart.

## Dashboard 2 — Geographic Distribution of Emissions

### Purpose

To show how emissions are distributed across countries, and how total emissions distribution differs from per-capita emissions distribution geographically.

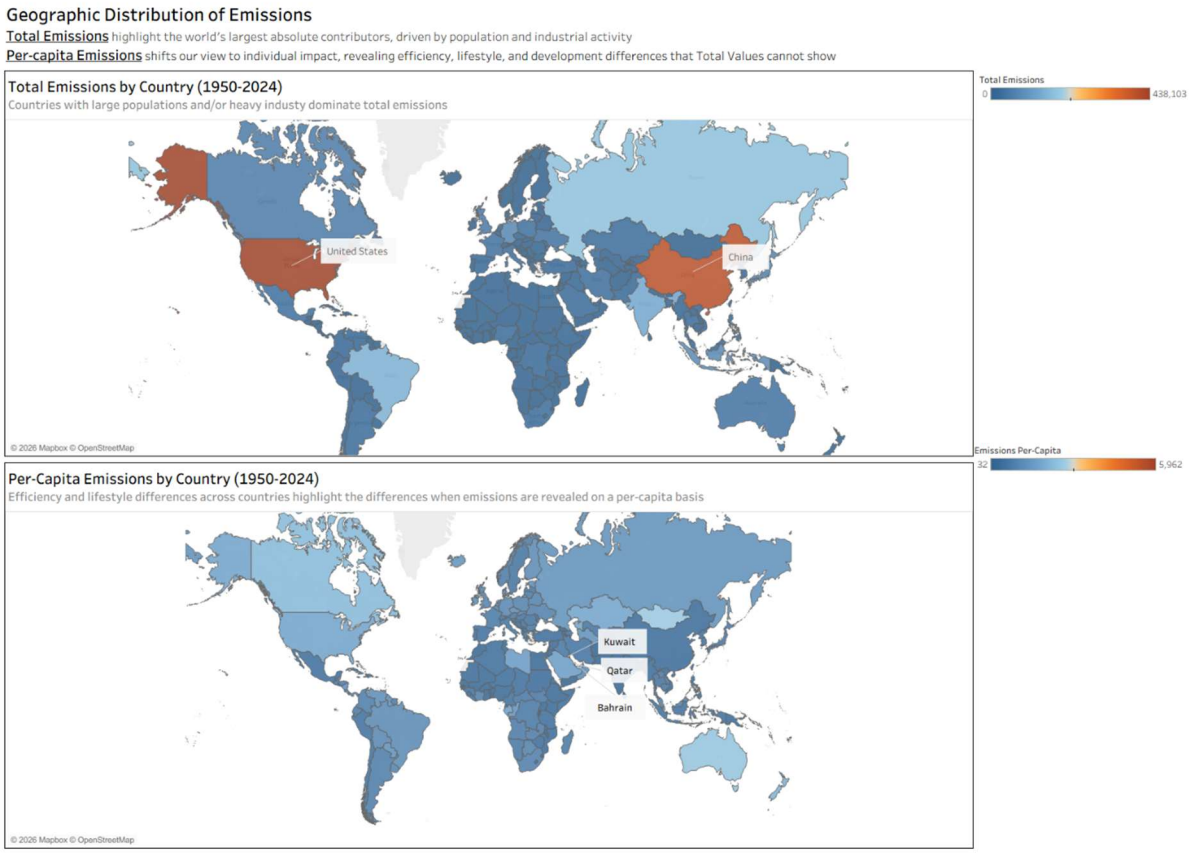
### Charts

1. **Total emissions are concentrated in a small number of large, industrialized, and populous countries**. Major emitters highlighted are high population or high industrial output countries such as the United States, China, and India.
2. **Per-capita emissions show a different set of countries.** Those countries have smaller populations but carbon/ fossil fuel-based economies, for example, Gulf states.

### Interpretation

These maps demonstrate that use of the term “large emitters” depends on what metric is used. There is a contrast between these two key types. Total emissions reflect global impact, where per-capita emissions reflect economic structure of the country and/ or region.

**Figure 2.**



### Effectiveness of Visuals

The Total Emissions filled map highlights the geographic concentration of total emissions, and scale differences between countries is clearly visible. Large emitters stand out visually even with no added labels. Global inequality regarding emissions is clearly seen. The map also shows that total emissions are clustered in a small number of high-population, high-industrialized countries.

When the Per-Capita filled map is presented, the emissions story changes to individual energy use, such as fossil fuels. These countries do not appear to be major emitters initially, but they are high consumption countries. High per-capita emitters must be included in any conversation when responsible climate policies are discussed.

## Dashboard 3 — Country-Level Profiles: Two Distinct Emissions Patterns Emerge

### Purpose

To compare the structure of global emissions at two points in time, 1950 and 2024, and to highlight how the distribution of emitters has shifted.

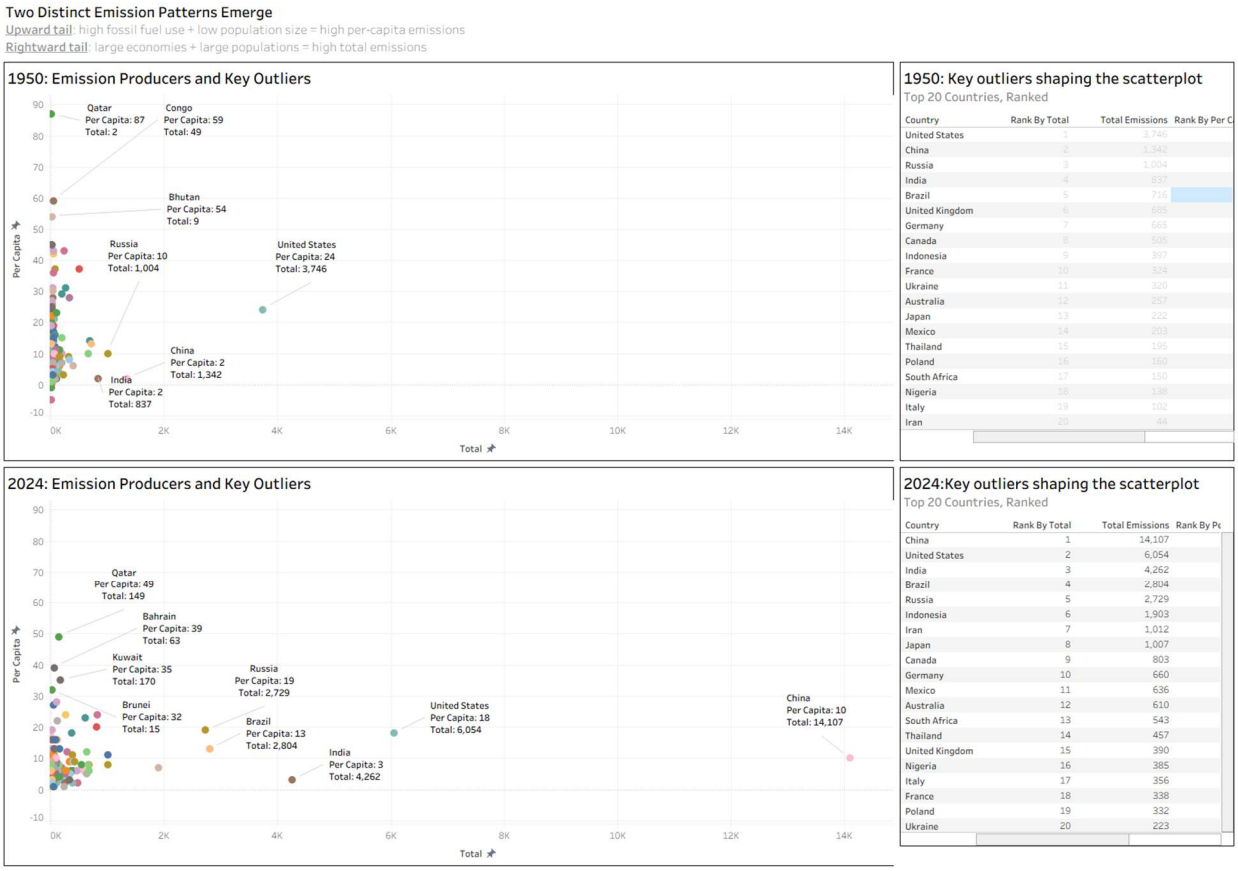
### Charts and Tables

1. **The 1950 scatterplot shows a world dominated by a few industrialized emitters**. Both total emissions and per-capita emissions were concentrated among a small number of countries with industrial economies.
2. **The scatterplot for 2024 shows a more complex landscape where rapidly developing economies moved into the high total emissions space and high-income countries remained elevated, per-capita**. The shape of this scatterplot spreads outward in both directions, reflecting global economic diversification.
3. **The accompanying tables clarify which countries are contained in the "tails" of the scatterplots**. Outliers, those with unusually high total emissions, per-capita emissions, or both, are more easily identified.

### Interpretation

Between 1950 and 2024 there were significant changes in many global emitters. Development, population growth, and energy use changes, as well as lack of emissions policy legislations in many countries, are reflected in the visible spreads for both total and per-capita emissions across countries.

**Figure 3.**




### Effectiveness of Visuals

The scatterplots show the change in distribution over time. The latest data from 2024 shows expanded rightward and upward tails in the distribution compared to the 1950 data, the earliest data in the dataset. Outliers are clearly visible, and one may compare total and per-capita emissions nearly simultaneously and having the oldest and most recent data on the same page strengthens the comparison of emissions across the 74-year time span.

The companion tables help to identify the scatterplots outliers, thus identifying which countries drive global patterns.

## Dashboard 4 — How Global Emissions Shifted Since 1950

### Purpose

To quantify which countries changed the most, positively or negatively, in both total and per-capita emissions between 1950 and 2024.

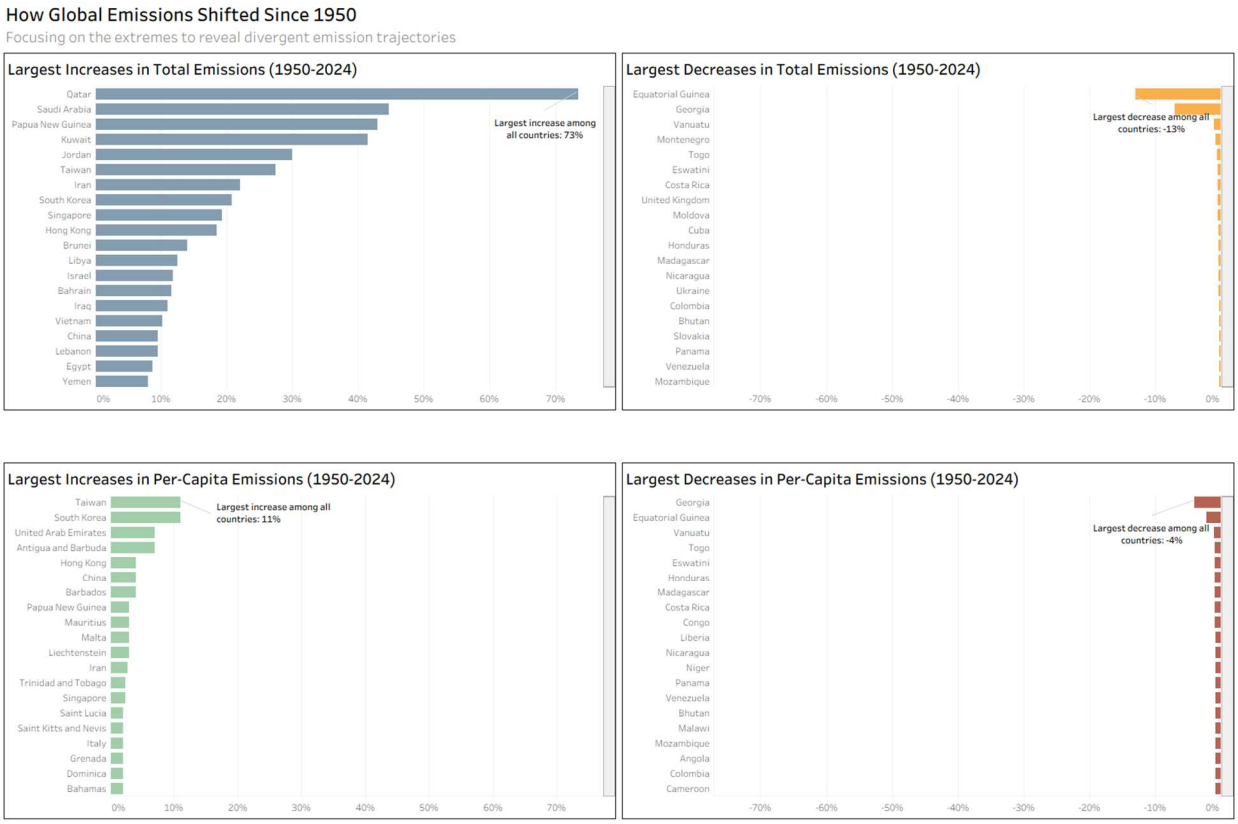
**Charts**

1. **Some countries show a dramatic increase in total emissions**. Often, these correspond to rapid industrialization, population growth, or shifts in energy resources such as fossil fuels, which are carbon intensive.
2. **Other countries show substantial decreases in total emissions**. Declines typically are reflected in changes in energy resources, such as moving away from fossil fuels or policy interventions to slow the emissions rate.
3. **Per-capita changes are a bit different**. Rising populations in some countries may show large total emissions increases but, for stable or declining per-capita emissions changes, the opposite may also be true. Countries with stable populations but have economies based more on fossil fuel consumption may show larger per-capita emissions increases.

**Interpretation**

The countries undergoing the most significant changes, upward and downward, are shown on this dashboard. It shows clear changes in many countries in 2024 that were not apparent in 1950.

**Figure 4.**




### Effectiveness of Visuals

The bar charts make patterns immediately clear as countries are ordered by rank. Magnitude is readily apparent in the bar charts, as the scale for each is the same, making comparisons of countries, ranges, and the increases/ decreases in emissions easy to communicate. The diverging bar format makes both positive and negative changes visually easy to see.

When I constructed the diverging bar chart with the center at zero, and I divided the data into positive and negative around that zero, this created a better visual than the change bar chart that showed total change in one direction. Therefore, I discarded that visual, and I did not use it, as it was not nearly effective as the ones presented for Dashboard 4.

## Summary

The structure of global emissions has shifted from 1950 to 2024, with emerging economies playing a larger role. Some countries made significant progress in emissions output, while

others experienced large increases in emissions, which shows the diversity of nations among key areas.

Global emissions have risen substantially, but per-capita emissions have not grown at the same rate. Emissions are unevenly distributed between countries, and care must be taken when discussing emissions rates that the metric used in those discussions is considered, whether total emissions or per-capita emissions.